

Md Mobarak Karim

Optical Systems Engineer | Optical Engineer

Houston, TX, USA | mkarim@uh.edu | LinkedIn | Portfolio | Google Scholar

Professional Summary

Optical Systems Engineer and Optical Engineer with a Ph.D. in Biomedical Engineering and 6+ years of experience designing, integrating, aligning, validating, and troubleshooting advanced optical imaging systems. Expertise in OCT, light-sheet fluorescence microscopy, two-photon microscopy, laser scanning, optical metrology, system characterization, and quantitative image analysis. Proficient in Zemax, MATLAB, Python, LabVIEW, and cross-functional optical system development; authorized to work in the United States without employer sponsorship.

Technical Skills

Optical Systems	Swept-source OCT, spectral-domain OCT, OCM, one-photon LSFM, two-photon LSFM, multiphoton microscopy, confocal microscopy, fluorescence microscopy, bright-field imaging, laser-scanning microscopy
Optical Engineering	Optical alignment, Gaussian beam propagation, beam expanders, 4f relays, scan/tube lenses, galvo integration, fiber optics, polarization control, interferometry, optical component qualification, Zemax, Optiland
System Characterization	Axial/lateral resolution, PSF, MTF, SNR, CNR, sensitivity, roll-off, field of view, depth of field, aberration analysis, image-quality evaluation, repeatability, calibration, design verification
Programming and Analysis	MATLAB, Python, C/C++, LabVIEW, ImageJ/Fiji, Imaris, Amira, 3D reconstruction, segmentation, co-registration, attenuation mapping, statistical analysis, GUI development
Engineering Practice	System integration, root-cause analysis, troubleshooting, process optimization, SOP development, quality-control workflows, technical documentation, vendor coordination, multidisciplinary leadership

Professional and Research Experience

Graduate Research Assistant - Multimodal Optical Systems

Jan 2021–May 2026

Biomedical Optics Laboratory, University of Houston

Houston, TX

- Designed and integrated a co-registered multimodal imaging platform combining a **1051-nm swept-source OCT system** (109-nm bandwidth, 100-kHz sweep rate) with **one-photon and two-photon LSFM** for simultaneous structural and molecular imaging of mouse and zebrafish embryos.
- Achieved approximately **7–7.6 micrometer OCT axial resolution**, **15 micrometer lateral resolution**, approximately **100 dB sensitivity**, and approximately **6.9 dB sensitivity roll-off over 4.13 mm**; characterized system resolution, sensitivity, roll-off, field of view, image contrast, and repeatability.
- Engineered beam delivery, polarization control, galvo scanning, scan-lens interfaces, acquisition timing, camera/laser synchronization, and sample positioning for stable multimodal 3D imaging.
- Demonstrated one-photon fluorescence penetration of approximately **250 micrometers** and two-photon visualization beyond **800 micrometers** in scattering embryonic tissue while OCT provided label-free whole-embryo morphology.
- Developed quantitative OCT algorithms for depth-resolved attenuation-coefficient estimation with noise subtraction, confocal correction, and sensitivity-roll-off compensation; validated the method using multilayer TiO₂ phantoms.
- Built MATLAB and Python pipelines for OCT reconstruction, 3D visualization, multimodal co-registration, morphometry, image-quality analysis, automated quality control, and standardized data processing.
- Developed cost-effective (300-500 USD) imaging incubator for the OCT-LS system.
- Mentored **three Ph.D. students and several undergraduate researchers** in OCT/LSFM system alignment, optical troubleshooting, image acquisition, MATLAB/Python analysis, experimental design, and technical presentation.
- Led optical hardware development, software integration, biological imaging, data analysis, manuscript preparation, and coordination across multidisciplinary research teams.

Graduate Research Assistant - Optical Design and Two-Photon Microscopy

Aug 2018–Dec 2020

3D Biomedical Imaging and Technology Laboratory, GIST

Gwangju, South Korea

- Designed and optimized broadband scan and tube lenses for laser-scanning microscopy using **Zemax OpticStudio**. Evaluated **wavefront error**, **Strehl ratio**, **aberrations**, **field of view**, **telecentricity**, and **F-theta distortion** across 680–1600 nm.
- Developed four optimized scan lenses and one tube lens, reducing **F-theta distortion from 13.00 to 1.94** and **telecentric distortion from 3.52 to 0.4367**.

- Supported development and qualification of a two-photon laser-scanning microscope through optical-component evaluation, alignment, testing, measured-performance analysis, and structured troubleshooting.

Education

Ph.D., Biomedical Engineering

University of Houston

Advisor: Dr. Kirill V. Larin. Dissertation: *Co-registered optical coherence tomography and light-sheet fluorescence microscopy for structural and molecular imaging of embryonic development.*

Jan 2021–May 2026

Houston, TX

M.S., Biomedical Science and Engineering

Gwangju Institute of Science and Technology

Thesis: *Analysis of an achromatic doublet as a scan lens for laser-scanning microscopy.*

Aug 2018–Dec 2020

Gwangju, South Korea

B.S., Electrical and Electronic Engineering

International Islamic University Chittagong

Aug 2013–Aug 2017

Chittagong, Bangladesh

Advanced Training

- Multiscale Biophotonics Training Program, Beckman Laser Institute, **University of California, Irvine**, 2024 - NIH-supported training in fluorescence lifetime imaging, nonlinear microscopy, spectroscopy, and translational biophotonics.
- CLIMB Biophotonics Summer School, Beckman Institute, **University of Illinois Urbana-Champaign**, 2023 - hands-on OCT/OCM system construction, reconstruction, and quantitative imaging.
- Siegman International School on Lasers, Poland**, 2022 - laser physics, ultrafast optics, nonlinear optics, and advanced photonic systems.

Selected Publications

- Md Mobarak Karim**, et al. “Multimodal optical coherence tomography and two-photon light-sheet fluorescence microscopy for embryo imaging.” *Journal of Biomedical Optics* (2025). Developed a co-registered structural-molecular platform integrating OCT and two-photon LSFM for intact-embryo imaging.
- Md Mobarak Karim**, et al. “Depth-resolved attenuation coefficient estimation during embryonic brain development.” *Journal of Biophotonics* (2025). Established quantitative OCT attenuation mapping for region-specific assessment of murine brain development.
- Leah A. Lewis, **Md Mobarak Karim**, et al. “Multimodal Optical Imaging for the Assessment of the Teratogenic Effects of Embryonic Ethanol Exposure on Zebrafish Development.” *Journal of Biomedical Optics* (2026). Integrated OCT and LSFM to quantify morphological and vascular developmental-toxicity phenotypes.
- James L. Klein, **Md Mobarak Karim**, Christian Zevallos-Delgado, Manmohan Singh, Salavat R. Aglyamov, and Kirill V. Larin. “Tartrazine-Based Optical Clearing for Monitoring Embryogenesis with Optical Coherence Tomography.” Submitted to *Optics Letters*. Demonstrated approximately twofold deeper OCT imaging of murine embryos using tartrazine-based optical clearing.
- Md Mobarak Karim**, et al. “Long-term imaging of zebrafish embryonic development with multimodal optical coherence tomography and light-sheet fluorescence microscopy.” Manuscript in preparation.

Awards and Honors

2025	Cover image featured in <i>Journal of Biomedical Optics</i>
2024	NIH Travel Grant, UCI Multiscale Biophotonics Training Program
2022	Zuegel Family Scholarship, Siegman International School on Lasers
2022	OPTICA Siegman International School on Lasers Grant
2022	SPIE Photonics West Travel Grant
2020	Outstanding Poster Award, Biomedical Research Meeting, South Korea
2018–2020	Korean Government Scholarship for M.S. study at GIST
2013–2017	Ungraduate Merit Award and Scholarship, Dept. of EEE, IIUC

Professional Service and Leadership

- Reviewer for biomedical optics and photonics journals.
- Senior Judge, Texas Science and Engineering Fair - Biomedical Division (2022).
- Executive Member, University of Houston SPIE Student Chapter (2021–Present).
- Member, OPTICA (formerly the Optical Society of America) (2021–Present).
- Founder and former executive member, Bangladesh Robotics Foundation.
- Troop Leader, Bangladesh Red Crescent Youth; Leadership Graduate, BYLC Art of Leadership Program (2016).